THE GLUEING FUNCTOR RESTRICTS TO EQUIVALENCES ON AFFINE-OVER-X, CLOSED IMMERSIONS, AND FINITE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2325-2341 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2343-2373, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 21 of queue_spaces-pushouts.json (pb=1398, stmt 2325-2341, proof 2343-2373).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0AF3: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2319–2321. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AF1: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2272–2276. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AF0: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2123–2127. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AEV: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2046–2060. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0420: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2373–2373. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0426: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2373–2373. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 081W: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2373–2373. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 081V: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2373–2373. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0AEV the functor (0AF3) restricts to an equivalence*

(1) *from the category of algebraic spaces affine over X to the full subcategory of Spaces$(Y \to X, Z)$ consisting of $(U' \leftarrow V' \to Y')$ with $U' \to U$, $V' \to V$, and $Y' \to Y$ affine,*
(2) *from the category of closed immersions $X' \to X$ to the full subcategory of Spaces$(Y \to X, Z)$ consisting of $(U' \leftarrow V' \to Y')$ with $U' \to U$, $V' \to V$, and $Y' \to Y$ closed immersions, and*
(3) *same statement as in (2) for finite morphisms.*

Proof. The category of algebraic spaces affine over X is equivalent to the category of quasi-coherent sheaves $\mathcal{A}$ of $\mathcal{O}_X$-algebras. The full subcategory of *Spaces*$(Y \to X, Z)$ consisting of $(U' \leftarrow V' \to Y')$ with $U' \to U$, $V' \to V$, and $Y' \to Y$ affine is equivalent to the category of algebra objects of $QCoh(Y \to X, Z)$. In both cases this follows from Morphisms of Spaces, Lemma 081V with quasi-inverse given by the relative spectrum construction (Morphisms of Spaces, Definition 081W) which commutes with arbitrary base change. Thus part (1) of the lemma follows from Proposition 0AF0.

Fully faithfulness in part (2) follows from part (1). For essential surjectivity, we reduce by part (1) to proving that $X' \to X$ is a closed immersion if and only if both $U \times_X X' \to U$ and $Y \times_X X' \to Y$ are closed immersions. By Lemma 0AF1 $\{U \to X, Y \to X\}$ can be refined by an fpqc covering. Hence the result follows from Descent on Spaces, Lemma 0420.

For (3) use the argument proving (2) and Descent on Spaces, Lemma 0426. □